

实验一（HW1）基础代码简述：

```
1 print('hello!')
2
3 # PyTorch
4 import torch
5 import torch.nn as nn
6 from torch.utils.data import Dataset, DataLoader
7
8 # For data preprocess
9 import numpy as np
10 import csv
11 import os
12
13 # For plotting
14 import matplotlib.pyplot as plt
15 from matplotlib.pyplot import figure
```

导入需要的包

```
26 tr_path = 'covid.train.csv' # path to training data
27 tt_path = 'covid.test.csv' # path to testing data
```

设置训练集和测试集路径

```
89         if not target_only:
90             feats = list(range(93))
91         else:
92             # TODO: Using 40 states & 2 tested_positive features (indices = 57 & 75)
93             pass
```

特征选择，if 条件下是 93 个特征（全特征），可以在 else 后选择自己想要的特征，此时需要更改代码 251 行的 target_only 为 True

```
107         if mode == 'train':
108             indices = [i for i in range(len(data)) if i % 10 != 0]
109         elif mode == 'dev':
110             indices = [i for i in range(len(data)) if i % 10 == 0]
```

划分训练集和验证集，可以更改划分方式

```
117         self.data[:, 40:] = \
118             (self.data[:, 40:] - self.data[:, 40:].mean(dim=0, keepdim=True)) \
119             / self.data[:, 40:].std(dim=0, keepdim=True)
120
121         self.dim = self.data.shape[1]
```

数据归一化，可更改归一化方式

```
150 class NeuralNet(nn.Module):
151     ''' A simple fully-connected deep neural network '''
152     def __init__(self, input_dim):
153         super(NeuralNet, self).__init__()
154
155         # Define your neural network here
156         # TODO: How to modify this model to achieve better performance?
157         self.net = nn.Sequential(
158             nn.Linear(input_dim, 64),
159             nn.ReLU(),
160             # nn.LeakyReLU(0.1),
161             nn.Linear(64, 1)
162         )
163
164         # Mean squared error loss
165         self.criterion = nn.MSELoss(reduction='mean')
```

模型结构部分，可以在这里修改层数等基本结构

```
171     def cal_loss(self, pred, target):  
172         ''' Calculate loss '''  
173         # TODO: you may implement L2 regularization here  
174         return self.criterion(pred, target)
```

Loss 计算部分，可更改 loss 计算方式

```
254     config = {  
255         'n_epochs': 3000,           # maximum number of epochs  
256         'batch_size': 270,         # mini-batch size for dataloader  
257         'optimizer': 'SGD',         # optimization algorithm (optimizer in torch.optim)  
258         'optim_hparas': {           # hyper-parameters for the optimizer (depends on which optimizer you are using)  
259             'lr': 0.001,             # learning rate of SGD  
260             'momentum': 0.9          # momentum for SGD  
261         },  
262         'early_stop': 300,          # early stopping epochs (the number epochs since your model's last improvement)  
263         'save_path': 'models/model.pth' # your model will be saved here
```

超参数设置部分，调参在这里